DETAILED TECHNICAL STUDY NOTES

# ReDiffDet

## Rotation-Equivariant Diffusion Model for Oriented Object Detection

Based on the CVPR 2025 paper by Jiaqi Zhao et al.

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Mathematical foundations, derivations,
architecture, algorithms, experiments, and limitations

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Part 1

Problem formulation, mathematical foundations,
Gaussian representation, and forward diffusion

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Paper: Jiaqi Zhao et al. · CVPR 2025 · 11 pages

We will study this paper from the ground up, including the mathematical foundations, every major equation, the derivations and proofs of rotation equivariance, the architecture, the training and inference algorithms, the experiments, and the limitations. The goal is to understand why ReDiffDet works, not merely memorize its architecture.

The explanation will follow the paper's actual methodology, with additional mathematical steps wherever they help make the reasoning clear. The main reference is the uploaded paper, particularly its Gaussian-modeling section on page 3, diffusion and equivariance proofs on page 4, algorithms on page 5, and experiments on pages 6–9.

ReDiffDet_Rotation-equivariant_Diffusion_Model_for_Oriented_Object_Detection(4).pdf

## 0.1 1. First understand the problem the paper solves

### 0.1.1 1.1 What is oriented object detection?

Suppose we are given an aerial image containing ships, aircraft, bridges, vehicles, and buildings. We want a model to predict:

1. Where each object is located.
2. How large it is.
3. The orientation at which it lies.
4. Which category it belongs to.

A conventional horizontal object detector represents an object using an axis-aligned rectangle. An oriented object detector instead predicts a rotated rectangle that follows the object's actual orientation.

Ship Detection under Complex Backgrounds Based on Accurate Rotated Anchor Boxes from Paired Semantic Segmentation

Horizontal bounding box

The rectangle remains horizontal, even if the ship is diagonal.

Horizontal bounding box detection and oriented bounding box detection... | Download Scientific Diagram

Oriented bounding box

The rectangle follows the ship's direction more closely.

Consider a long ship tilted by 35°. A horizontal rectangle must include additional water around the ship. When several ships are parked close together, these extra regions can cause overlapping boxes and inaccurate localization.

This is especially problematic for aerial images because:

- Objects can point in almost any direction.
- Many objects are tightly packed together.

- Ships, aircraft and bridges often have extreme aspect ratios.
- Objects may occupy only a few pixels.
- Rotating an object changes how its visual features appear on an ordinary image grid.

### 0.1.2 1.2 The mathematical representation of a rotated box

ReDiffDet represents an oriented box as

$$b = (c_x, c_y, w, h, a),$$

where:

- $c_x, c_y$: center coordinates.
- $w$: width along the box's local width axis.
- $h$: height along its local height axis.
- $a$: rotation angle.

The five values describe the box's position, size and orientation.

For a horizontal detector, the representation usually has four values:

$$b_\text{horizontal} = (c_x, c_y, w, h).$$

The angle $a$ introduces a fundamental challenge: the geometry of the box must rotate consistently with the object and its visual features.

A detector that learns only from axis-aligned features may predict the wrong orientation or fail to align the object's long axis with its bounding box.

### 0.1.3 1.3 Why diffusion models are relevant

A conventional detector often directly predicts boxes from image features. A diffusion detector takes a different approach: it treats detection as a process of progressively refining noisy candidate boxes.

The basic idea is:

$$\text{Random/noisy candidates} \longrightarrow \text{less noisy candidates} \longrightarrow \text{object boxes}.$$

This is analogous to generative diffusion models that transform noise into structured data. The difference is that ReDiffDet generates object representations, rather than images.

The paper identifies three problems when applying diffusion directly to oriented object detection:

| Problem | Why it matters |
|---|---|
| Rotation equivariance | Predictions should rotate consistently when the image rotates. |
| Poorly controlled initial boxes | Random candidates may miss small, elongated or densely packed objects. |
| Feature misalignment | Axis-aligned feature maps do not naturally match rotated object regions. |

ReDiffDet addresses these three issues with three interconnected ideas:

1. Represent oriented boxes as 2D Gaussian distributions.
2. Use a conditional encoder to provide location-aware candidate boxes.
3. Use an aligned decoder that extracts features from the rotated regions and refines the candidates through diffusion.

The most important conceptual shift is that ReDiffDet does not diffuse the five box parameters independently. It diffuses a geometric distribution representing the box.

## 0.2 2. The mathematical foundations you need

Before studying the architecture, we need to understand five concepts:

- Probability distributions and Gaussian random variables.
- Covariance matrices and eigenvectors.
- Rotation matrices and the group $SO(2)$.
- Diffusion's forward and reverse processes.
- Equivariance versus invariance.

These concepts are directly connected to the equations used in the paper.

### 0.2.1 2.1 The multivariate Gaussian distribution

A one-dimensional Gaussian random variable is described by

$$x \sim \mathcal{N}(\mu, \sigma^2),$$

where $\mu$ is its mean and $\sigma^2$ is its variance.

The probability density function is

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

The mean tells us where the distribution is centered, while the variance describes its spread.

For two-dimensional data, the corresponding distribution is

$$\mathbf{z} = \begin{bmatrix} x \\ y \end{bmatrix} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma),$$

with mean vector

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}$$

and covariance matrix

$$\Sigma = \begin{bmatrix} \sigma_x^2 & \sigma_xy \\ \sigma_xy & \sigma_y^2 \end{bmatrix}.$$

Here, $\sigma_xy$ is the covariance between the horizontal and vertical coordinates. It measures whether deviations in $x$ and $y$ tend to occur together.

A covariance matrix is symmetric and positive semidefinite. For a nondegenerate 2D Gaussian, it is positive definite, so its determinant is positive and its inverse exists.

### 0.2.2 2.2 What does covariance mean geometrically?

Imagine sampling thousands of points from a 2D Gaussian and plotting them.

GMM Appendix B: Covariance Matrix Structure in Multivariate Gaussians – Tools for Data Science

Case A: Equal spread in every direction

$$\Sigma = \sigma^2 I$$

The distribution has circular contours. There is no preferred orientation.

Eccentricity of bivariate normal level sets

Case B: Different spreads in different directions

$$\Sigma = \begin{bmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{bmatrix}, \qquad \sigma_x \neq \sigma_y$$

The contours are elliptical. The distribution has a preferred direction.

GSB 518 Handouts - 15 Joint Normal Distributions

Case C: The ellipse is rotated

The covariance matrix has off-diagonal terms in the global coordinate system. Its eigenvectors indicate the ellipse's principal directions.

This is the key geometric insight behind ReDiffDet.

A long, narrow object naturally corresponds to a long, narrow Gaussian ellipse. The ellipse's center represents the box center, its principal directions represent the box axes, and its spread represents the box dimensions.

The model can therefore operate on a distribution that explicitly encodes orientation.

### 0.2.3 2.3 Rotation matrices

A rotation matrix describes how coordinates change when a point is rotated.

For a counterclockwise rotation through angle $\phi$, define

$$Q_\phi = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}.$$

For a point

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix},$$

its rotated coordinates are

$$\mathbf{x}' = Q_\phi \mathbf{x}.$$

Expanding this matrix multiplication gives

$$\begin{aligned} x' &= x\cos\phi - y\sin\phi, \\ y' &= x\sin\phi + y\cos\phi. \end{aligned}$$

These equations rotate the point without changing its distance from the rotation origin.

A fundamental property is

$$Q_\phi^\top Q_\phi = I.$$

Proof. Write $c = \cos\phi$ and $s = \sin\phi$. Then

$$Q_\phi^\top Q_\phi = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} c & -s \\ s & c \end{bmatrix}.$$

Multiplying gives

$$Q_\phi^\top Q_\phi = \begin{bmatrix} c^2 + s^2 & -cs + sc \\ -sc + cs & s^2 + c^2 \end{bmatrix}.$$

Using $c^2 + s^2 = 1$,

$$\boxed{Q_\phi^\top Q_\phi = I.}$$

Therefore, rotation preserves lengths and angles.

The determinant is also

$$\det Q_\phi = \cos^2\phi + \sin^2\phi = 1.$$

Consequently, $Q_\phi$ belongs to the special orthogonal group $SO(2)$, the set of all two-dimensional rotations.

### 0.2.4 2.4 What is $SO(2)$, and why does the paper use it?

The paper defines its transformations using $SO(2)$.

Formally,

$$SO(2) = \left\{ Q \in \mathbb{R}^{2\times 2} : Q^\top Q = I,\ \det Q = 1 \right\}.$$

This set contains every rotation matrix in two dimensions. It excludes reflections because a reflection reverses orientation and has determinant $-1$.

The group has three useful properties:

1. The product of two rotations is another rotation.
2. The identity matrix represents zero rotation.
3. Every rotation has an inverse, which is a rotation in the opposite direction.

For example,

$$Q_\phi_1 Q_\phi_2 = Q_\phi_1 + \phi_2.$$

Thus, rotating an image by 20°, then by 30°, is equivalent to rotating it by 50°.

The reason this group matters is that an oriented detector should respond predictably to these transformations.

There is one important distinction: the paper's formal proofs concern rotations in the modeled 2D coordinate system. Applying the same mathematics to an actual raster image also involves image interpolation, padding, discretization and boundary effects.

### 0.2.5 2.5 Equivariance versus invariance

These are two of the most important concepts in the paper.

Rotation equivariance means that transforming the input transforms the output in a corresponding way.

If $f$ is a detector and $Q$ rotates the image, the ideal relation is

$$\boxed{f(QI) = Qf(I).}$$

Here, $QI$ means the rotated image and $Qf(I)$ means the original predictions transformed by the same rotation.

For example, if a ship's predicted box has orientation 25°, rotating the image by 40° should produce a corresponding box with orientation 65°, subject to the angle convention and equivalent box representations.

The prediction does not remain unchanged. It rotates with the image.

Rotation invariance, by contrast, means that the output remains unchanged:

$$\boxed{g(QI) = g(I).}$$

For example, a classification feature might identify a ship as a ship regardless of whether it points north or east.

| Property | What changes after rotating the input? | Example |
|---|---|---|
| Equivariance | Output transforms with the input | A bounding box rotates with its object |
| Invariance | Output remains unchanged | An object's category remains "ship" |

Why does ReDiffDet need both ideas?

- Its box predictions must be equivariant: their geometry should follow object rotation.
- Its Gaussian noise distribution must be invariant: rotating isotropic Gaussian noise should not change its probability law.
- Its feature extraction and rotated-region alignment mechanisms help connect these two properties.

The paper's Equations (1) and (2) formalize these definitions on page 3.

## 0.3 3. The central innovation: representing an oriented box as a 2D Gaussian

We now arrive at the core of ReDiffDet.

### 0.3.1 3.1 Converting a box into a Gaussian

Suppose the oriented box is

$$b = (c_x, c_y, w, h, a).$$

ReDiffDet converts it into a Gaussian distribution

$$\boxed{z_0 \sim \mathcal{N}(\mu, \Sigma),}$$

where

$$\boxed{\mu = \begin{bmatrix} c_x \\ c_y \end{bmatrix}, \qquad \Sigma = R\Lambda R^\top.}$$

The rotation matrix for the box angle $a$ is

$$R = \begin{bmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{bmatrix}.$$

The diagonal matrix is

$$\Lambda = \begin{bmatrix} (w/m)^2 & 0 \\ 0 & (h/m)^2 \end{bmatrix},$$

where $m$ is a scaling factor.

These are the equations given in Equation (3) of the paper.

Let's understand every component.

- $\mu$ is the center of the object.
- $R$ specifies the orientation of the ellipse.
- $\Lambda$ specifies the spread along the ellipse's two principal axes.
- $m$ controls the amount of spread relative to the box dimensions.
- $\Sigma$ combines orientation and spread into one covariance matrix.

The use of $w/m$ and $h/m$, rather than $w$ and $h$ directly, is a design choice. It controls how the sampled points are distributed around the box center.

### 0.3.2 3.2 Deriving the covariance matrix

Let's derive the expression

$$\Sigma = R\Lambda R^\top.$$

Start with a random vector in the box's local coordinate system:

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \qquad X \sim \mathcal{N}(0, \Lambda).$$

Its covariance is

$$\mathrm{Cov}(X) = \Lambda.$$

Rotate it and translate it to the box center:

$$Z = \mu + RX.$$

The mean is

$$\begin{aligned} E[Z] &= E[\mu + RX] \\ &= \mu + RE[X] \\ &= \mu, \end{aligned}$$

because $E[X] = 0$.

Now derive the covariance. By definition,

$$\mathrm{Cov}(Z) = E[(Z - E[Z])(Z - E[Z])^\top].$$

Since $E[Z] = \mu$,

$$Z - E[Z] = RX.$$

Therefore,

$$\begin{aligned} \mathrm{Cov}(Z) &= E[(RX)(RX)^\top] \\ &= E[RXX^\top R^\top] \\ &= RE[XX^\top]R^\top \\ &= R\Lambda R^\top. \end{aligned}$$

Thus,

$$\boxed{\Sigma = R\Lambda R^\top.}$$

This derivation is important because it explains precisely how a box's orientation and dimensions become part of a Gaussian distribution.

### 0.3.3 3.3 Expanding the covariance into scalar terms

For deeper understanding, let

$$A = (w/m)^2, \qquad B = (h/m)^2,$$

and abbreviate $c = \cos a,\ s = \sin a$.

Then

$$\Sigma = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \begin{bmatrix} c & s \\ -s & c \end{bmatrix}.$$

First,

$$R\Lambda = \begin{bmatrix} cA & -sB \\ sA & cB \end{bmatrix}.$$

Multiplying by $R^\top$,

$$\boxed{\Sigma = \begin{bmatrix} Ac^2 + Bs^2 & (A-B)cs \\ (A-B)cs & As^2 + Bc^2 \end{bmatrix}.}$$

This expanded form reveals how the angle enters the covariance.

The off-diagonal covariance is

$$\Sigma_12 = (A - B)\sin a \cos a.$$

When the box is elongated and rotated, this term is generally nonzero. The horizontal and vertical coordinates become statistically correlated.

Special cases help build intuition:

- If $a = 0$, the covariance is diagonal.
- If $w = h$, the covariance is isotropic and independent of $a$.
- If $w \gg h$, the distribution becomes elongated along the box's width axis.

Notice that a square has no uniquely defined principal orientation from its covariance alone. That is a genuine geometric ambiguity, not a mathematical error.

#### 0.3.4 3.4 A numerical example

Suppose

$$c_x = 100, \quad c_y = 80, \quad w = 60, \quad h = 20, \quad a = 30^\circ, \quad m = 6.$$

Then

$$\mu = \begin{bmatrix} 100 \\ 80 \end{bmatrix},$$

and

$$\Lambda = \begin{bmatrix} (60/6)^2 & 0 \\ 0 & (20/6)^2 \end{bmatrix} = \begin{bmatrix} 100 & 0 \\ 0 & 11.11 \end{bmatrix}.$$

The rotation matrix is approximately

$$R = \begin{bmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{bmatrix}.$$

Substituting into $R\Lambda R^\top$ gives

$$\Sigma \approx \begin{bmatrix} 77.78 & 38.49 \\ 38.49 & 33.33 \end{bmatrix}.$$

The covariance is not diagonal because the ellipse is tilted. Its mean remains at $(100, 80)$.

This Gaussian is the mathematical object that ReDiffDet will corrupt with noise and subsequently denoise.

### 0.3.5 3.5 The probability density function

The paper next defines the Gaussian probability density in Equation (4):

$$p(z) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\left[-\frac{1}{2}(z-\mu)^\top \Sigma^{-1}(z-\mu)\right].$$

Here $z = (u, v)^\top$, and $|\Sigma|$ denotes the determinant.

The quadratic form

$$(z-\mu)^\top \Sigma^{-1}(z-\mu)$$

measures how far a point lies from the center relative to the Gaussian's covariance.

For an isotropic covariance, this is a scaled squared Euclidean distance. For an anisotropic covariance, distances along the narrow axis are penalized more strongly than equal-sized deviations along the broad axis.

The normalization factor ensures that the density integrates to one over the plane.

There is an important typographical distinction worth preserving when reading the paper: the Gaussian density in two dimensions has the normalization $(2\pi\sqrt{|\Sigma|})^{-1}$, equivalently $(2\pi)^{-1}|\Sigma|^{-1/2}$.

### 0.3.6 3.6 Why sample points from the Gaussian?

ReDiffDet does not merely create a Gaussian and then perform diffusion directly on its five box parameters. Its training algorithm samples points from the Gaussian and reconstructs the distribution from those points.

Suppose

$$P_i \sim \mathcal{N}(\mu, \Sigma), \qquad i = 1, \ldots, k.$$

The sample set is

$$P = P_1, P_2, \ldots, P_k.$$

The paper uses maximum likelihood estimation (MLE) to recover the distribution from these points.

The likelihood is

$$L(\mu, \Sigma) = \prod_i = 1^k p(P_i \mid \mu, \Sigma).$$

MLE chooses parameters maximizing this product:

$$\boxed{(\hat{\mu}, \hat{\Sigma}) = \arg\max_\mu, \Sigma \prod_i = 1^k p(P_i \mid \mu, \Sigma).}$$

This is the paper's Equation (5).

For independent samples from a nondegenerate multivariate Gaussian, the maximum-likelihood estimators are

$$\boxed{\hat{\mu} = \frac{1}{k} \sum_i = 1^k P_i}$$

and

$$\boxed{\hat{\Sigma}_\text{MLE} = \frac{1}{k} \sum_i = 1^k (P_i - \hat{\mu})(P_i - \hat{\mu})^\top.}$$

The denominator is $k$, not $k - 1$, because this is the maximum-likelihood covariance estimator rather than the usual unbiased sample covariance estimator.

Why does this matter? Because the model can perturb a finite set of points with noise, estimate their new distribution, and use that distribution as the current state of diffusion.

One practical qualification: with only a few points, the covariance estimate is noisy and may be rank deficient. This is a finite-sample issue, and the paper explicitly studies the number of samples in its ablation experiments.

### 0.3.7 3.7 Recovering a Gaussian from its covariance

After the distribution has been estimated, ReDiffDet must recover the oriented box.

The center is simply

$$\boxed{\begin{bmatrix} c_x \\ c_y \end{bmatrix} = \hat{\mu}.}$$

The remaining quantities come from the eigendecomposition of the covariance matrix.

For a symmetric covariance matrix,

$$\Sigma = R\Lambda R^\top,$$

where

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}.$$

The eigenvalues satisfy

$$\det(\Sigma - \lambda I) = 0.$$

For each eigenvalue, the associated eigenvector satisfies

$$(\Sigma - \lambda_i I) r_i = 0.$$

The normalized eigenvectors form the columns of $R$. The eigenvalues describe the variances along the principal directions.

If the width direction corresponds to $\lambda_1$, then, under the paper's Gaussian scaling convention,

$$w = m\sqrt{\lambda_1}, \qquad h = m\sqrt{\lambda_2}.$$

The orientation $a$ is obtained from the principal eigenvector, with the appropriate convention for which eigenvector represents width.

Why is eigendecomposition the right operation?

Because the covariance matrix is already expressed in global image coordinates. Its eigenvectors tell us the directions in which the Gaussian has its principal spreads, and its eigenvalues tell us how large those spreads are.

This closes the representation loop:

$$\boxed{\text{Oriented box} \longrightarrow \text{Gaussian} \longrightarrow \text{sample points} \longrightarrow \text{Gaussian} \longrightarrow \text{oriented box.}}$$

This conversion is the foundation of the entire method.

One subtlety: a Gaussian covariance does not uniquely distinguish an angle $a$ from $a + \pi$, because a rectangle's principal axis is undirected. Swapping width and height can also produce equivalent rectangle representations under a corresponding $90^\circ$ angle change. The implementation must follow a consistent oriented-box convention.

## 0.4 4. The diffusion process: how boxes become noise

We now have a representation that captures box geometry. The next question is:

How do we turn a meaningful Gaussian distribution into noise, and how can a neural network learn to reverse that process?

### 0.4.1 4.1 The forward diffusion process

In diffusion models, the forward process gradually corrupts a data sample by adding Gaussian noise.

ReDiffDet starts from a clean Gaussian representation $z_0$ of an oriented box. It then generates increasingly noisy states:

$$z_0 \to z_1 \to z_2 \to \cdots \to z_T.$$

The transition is defined by a Markov chain.

Let

$$\alpha_t = 1 - \beta_t,$$

where $\beta_t \in (0, 1)$ controls the amount of noise added at step $t$.

Define

$$\bar{\alpha}_t = \prod_i = 1^t \alpha_i.$$

The forward distribution can be written as

$$\boxed{q(z_t \mid z_0) = \mathcal{N}\left(z_t; \sqrt{\bar{\alpha}_t}, z_0, (1 - \bar{\alpha}_t)I\right).}$$

This is Equation (8) in the paper.

The equivalent sampling equation is

$$\boxed{z_t = \sqrt{\bar{\alpha}_t}, z_0 + \sqrt{1 - \bar{\alpha}_t}, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).}$$

Let's derive it carefully.

### 0.4.2 4.2 Deriving the closed-form forward equation

The one-step transition is

$$q(z_t \mid z_t - 1) = \mathcal{N}(\sqrt{\alpha_t} z_t - 1, \beta_t I).$$

Therefore, we can sample the next state using

$$z_t = \sqrt{\alpha_t} z_t - 1 + \sqrt{\beta_t} \epsilon_t,$$

where each $\epsilon_t$ is independent standard Gaussian noise.

At the first step,

$$z_1 = \sqrt{\alpha_1} z_0 + \sqrt{\beta_1} \epsilon_1.$$

At the second step,

$$z_2 = \sqrt{\alpha_2} z_1 + \sqrt{\beta_2} \epsilon_2.$$

Substitute the expression for $z_1$:

$$z_2 = \sqrt{\alpha_1 \alpha_2} z_0 + \sqrt{\alpha_2 \beta_1} \epsilon_1 + \sqrt{\beta_2} \epsilon_2.$$

The mean coefficient is

$$\sqrt{\alpha_1 \alpha_2} = \sqrt{\bar{\alpha}_2}.$$

Since independent Gaussian random variables combine into another Gaussian, the total noise variance is

$$\alpha_2 \beta_1 + \beta_2.$$

Using $\beta_1 = 1 - \alpha_1$,

$$\begin{aligned} \alpha_2 \beta_1 + \beta_2 &= \alpha_2 (1 - \alpha_1) + (1 - \alpha_2) \\ &= 1 - \alpha_1 \alpha_2 \\ &= 1 - \bar{\alpha}_2. \end{aligned}$$

The same reasoning extends to any timestep $t$. The resulting noise variance is

$$1 - \prod_i = 1^t \alpha_i = 1 - \bar{\alpha}_t.$$

Therefore,

$$z_t \sim \mathcal{N}\left(\sqrt{\bar{\alpha}_t} z_0, (1 - \bar{\alpha}_t) I\right),$$

as required.

This derivation explains why diffusion training can sample a noisy state directly at a randomly chosen timestep, rather than simulating every preceding step.

### 0.4.3 4.3 What happens as the timestep increases?

At early timesteps, $\bar{\alpha}_t$ is relatively large. The original representation contributes strongly to the state.

At later timesteps, $\bar{\alpha}_t$ becomes smaller, so the noise contribution increases.

$t = 0$: clean state

The Gaussian encodes the original box center, orientation and size.

Intermediate timestep

Noise disturbs the sampled points, changing the estimated center and covariance.

$t = T$: approximately pure noise

The original geometric signal is largely lost, leaving a state close to a standard Gaussian.

Strictly speaking, the standard-normal limit is approached when the schedule makes $\bar{\alpha}_T$ sufficiently close to zero. A finite diffusion process does not automatically produce an exact standard normal distribution for every possible clean-data distribution.

#### 0.4.4 4.4 What exactly is being diffused in ReDiffDet?

This is an important implementation detail.

The paper's Algorithm 1 does not simply take the five values $(c_x, c_y, w, h, a)$ and add independent noise to them.

Instead, it:

1. Converts ground-truth boxes into Gaussian distributions.
2. Samples points from those distributions.
3. Applies a coordinate transformation and signal scaling.
4. Adds Gaussian noise to the points using the diffusion equation.
5. Estimates a Gaussian from the corrupted points using MLE.
6. Converts the predicted distribution back into boxes for supervision.

This is why Gaussian modeling is more than a cosmetic representation change. It changes the space in which the diffusion process operates.